

WINDOWS SERVER EXAM REPORT

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Practical Exam Report

DHCP Server Configuration and Dynamic IP Assignment to Client.

Objective:

The objective of this practical was to configure a **Windows Server 2012** as a **DHCP server** and assign dynamic IP addresses to a **client machine (Windows 7)**.

Steps Followed:

1. Setting Up Windows Server 2012:

1. Installing Windows Server:

- First, I installed **Windows Server 2012** on the system.

2. Changing the Computer Name:

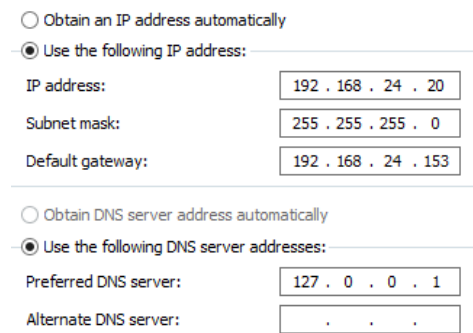
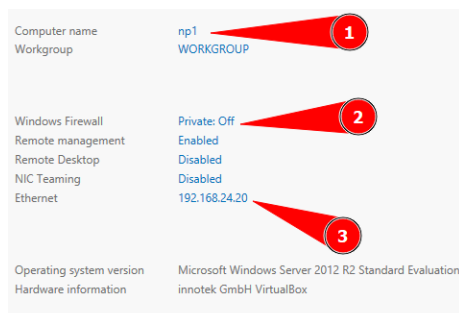
- I changed the **Computer Name** in the **Local Server** settings and restarted the server to apply the changes.

3. Disabling Windows Firewall:

- In the **Local Server Properties**, I disabled the **Windows Firewall** to avoid any issues with network communication.

4. Assigning Static IP to Server:

- In the **Ethernet Adapter** properties, I assigned a static **IPv4** address to the server:
 - IP Address:** 192.168.24.20
 - Subnet Mask:** 255.255.255.0
 - Default Gateway:** 192.168.24.153
 - DNS:** 127.0.0.1 (Local DNS)



2. Installing and Configuring DHCP Server:

1. Installing the DHCP Server Role:

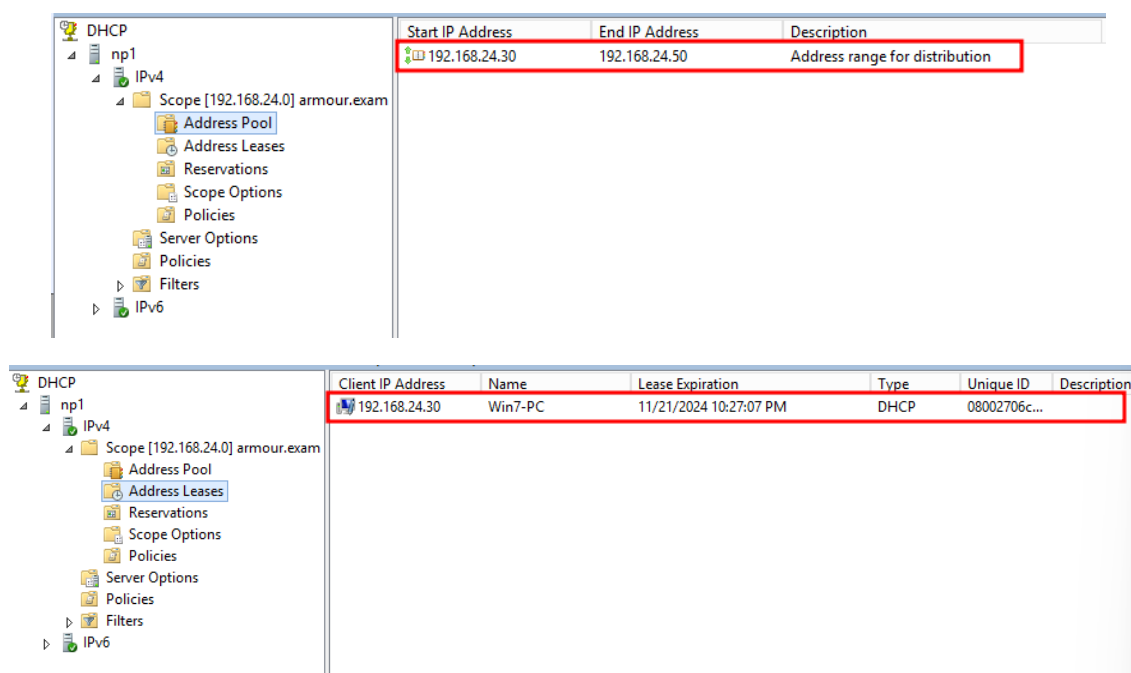
- I went to the **Manage** menu, selected **Add Roles and Features**, and installed the **DHCP Server** role.

2. Configuring the DHCP Server:

- After the DHCP server was installed, I opened the **DHCP Management Console**.
- Under **IPv4**, I created a **New Scope**:
 - **Scope Name**: armour.exam
 - **IP Range**: From 192.168.24.30 to 192.168.24.50
 - **Subnet Mask**: 255.255.255.0
 - **Lease Duration**: 3 Minutes.

3. Activating the DHCP Server:

- After creating the scope, I activated the **DHCP Server** to allow dynamic IP assignments to client machines.



3. Configuring the Client Machine (Windows 7):

1. Installing Windows 7 on Client:

- I installed **Windows 7** on the client machine.

2. Configuring Network Settings:

- On the client machine, I used the **Run command** `ncpa.cpl` to open **Network Connections**.
- I right-clicked the **Ethernet Adapter**, selected **Properties**, and unchecked **Internet Protocol Version 6 (IPv6)**. Then, I selected **Internet Protocol Version 4 (IPv4)**.
- I chose the option **Obtain an IP address automatically** and clicked **OK** to save the settings.

3. Verifying IP Address Assignment:

- The client machine automatically obtained an IP address from the **DHCP Server**. The client was assigned the IP **192.168.24.30**.
- I verified the IP configuration by running the `ipconfig` command in **Command Prompt** and confirmed the assigned IP address.

4. Verifying Internet Connectivity:

- I checked the **Internet Connectivity** on the client machine and confirmed that the internet was working and the machine was successfully connected to the server over the network.

```
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Win7>ping 192.168.24.20
Pinging 192.168.24.20 with 32 bytes of data:
Reply from 192.168.24.20: bytes=32 time<1ms TTL=128
Reply from 192.168.24.20: bytes=32 time<1ms TTL=128
Reply from 192.168.24.20: bytes=32 time<1ms TTL=128
Reply from 192.168.24.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.24.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Win7>
```

Conclusion:

In this practical, I successfully set up **Windows Server 2012** as a **DHCP Server**, configured a scope with an IP range from **192.168.24.30 to 192.168.24.50**, and assigned dynamic IP addresses to a **Windows 7 client machine**. The DHCP server assigned the first available IP address, **192.168.24.30**, to the client. The client machine was able to access the internet and was successfully connected to the server over the network.

Report on Applying Top 10 Group Policy Objects (GPOs) for the Sales Department Report

Step 1: Installing the AD DS Role

- First, I opened the **Add Roles and Features** wizard and selected the **Active Directory Domain Services (AD DS)** role.
- After that, I clicked **Next** and completed the installation of the AD DS role. Once installed, I proceeded to configure the server to function as a domain controller.

Step 2: Configuring AD DS

- After installing the AD DS role, I selected **Promote this server to a domain controller**.
- I chose the option **Add a new forest** and entered **Armour.exam** as the **Root domain name**.
- The **NetBIOS domain name** was automatically set to **ARMOUR**.
- I completed the configuration and restarted the server to finalize the setup of the domain controller.

Step 3: Creating Organizational Units (OU) and Users

- Next, I opened the **Active Directory Users and Computers** tool and created a new **Organizational Unit (OU)** named **Sales Department**.
- Inside this OU, I created **3 new users: Sale1, Sale2, and Sale3**.

Step 4: Creating and Applying Group Policies (GPO)

- Then, I opened the **Group Policy Management** tool and created a new **Group Policy Object (GPO)** for the **Sales Department**, naming it **Sale**.
- After configuring the GPO, I applied it to the required users and systems.

Step 5: Domain Joining on Client PC

- I then opened the **Windows 7 client PC**, changed its **Computer Name** from **win7-pc** to **win7**, and proceeded to join it to the **Armour.exam domain**.
 - **Username:** Sale1
 - **Password:** Sale@123
- After the domain join, I restarted the PC to ensure that the domain connection was established correctly.

Step 6: Verifying Connectivity

- After the client PC was successfully joined to the domain, I tested the network connectivity by **pinging** the server's **IP address**, which was successful.
- I also performed an **nslookup** command to verify DNS resolution, which also succeeded.

Step 7: Applying Group Policies

- I opened the **Group Policy Management Console (GPMC)** using the **gpmc.msc** command and applied the **10 policies** listed below, which are important for security and system settings. These policies were configured at their respective locations within the Group Policy Management tool.

Applied Group Policies:

1. **Control Panel Access Moderation**
 - **Path:** User Configuration → Administrative Templates → Control Panel
 - **Policy Name:** "Prohibit access to Control Panel and PC settings"
2. **Prevent Windows from Storing LAN Manager Hash**
 - **Path:** Computer Configuration → Windows Settings → Security Settings → Local Policies → Security Options
 - **Policy Name:** "Accounts: Limit local account use of blank passwords to console logon only"
3. **Control Access to Command Prompt**
 - **Path:** User Configuration → Administrative Templates → System
 - **Policy Name:** "Prevent access to the command prompt"
4. **Disable Forced System Restarts for Logged-On Users**
 - **Path:** Computer Configuration → Administrative Templates → Windows Components → Windows Update
 - **Policy Name:** "No auto-restart with logged on users for scheduled automatic updates installations"
5. **Disallow Removable Media Drives**
 - **Path:** Computer Configuration → Administrative Templates → System
 - **Policy Name:** "Removable Storage Access"
 - **Option:** "Deny all access to all Removable Storage classes"
6. **Restrict Software Installations**
 - **Path:** User Configuration → Administrative Templates → Windows Components → Windows Installer
 - **Policy Name:** "Disable Windows Installer"
7. **Disable Guest Account**
 - **Path:** Computer Configuration → Windows Settings → Security Settings → Local Policies → User Rights Assignment
 - **Policy Name:** "Accounts: Guest account status"
8. **Set Minimum Password Length**
 - **Path:** Computer Configuration → Windows Settings → Security Settings → Account Policies → Password Policy
 - **Policy Name:** "Minimum password length"
9. **Set Maximum Password Age**
 - **Path:** Computer Configuration → Windows Settings → Security Settings → Account Policies → Password Policy
 - **Policy Name:** "Maximum password age"
10. **Set Password Complexity Requirements**
 - **Path:** Computer Configuration → Windows Settings → Security Settings → Account Policies → Password Policy
 - **Policy Name:** "Password must meet complexity requirements"

Step 8: Applying the Policies

- After configuring the GPOs, I applied the policies by running the **gpupdate /force** command on the server, which ensured that the policies were pushed to the client PC and applied successfully.

Step 9: Final Verification

- Once the policies were applied, I verified that all settings were working as expected and that the system was secure according to the configured policies.

sale

ScopeDetailsSettingsDelegation

sale

Data collected on: 11/22/2024 1:55:55 AM

Computer Configuration (Enabled)

Policies

Windows Settings

Security Settings

Account Policies/Password Policy

Policy	Setting
Maximum password age	30 days
Minimum password age	29 days
Minimum password length	4 characters

Account Policies/Account Lockout Policy

Policy	Setting
Account lockout duration	30 minutes
Account lockout threshold	5 invalid logon attempts
Reset account lockout counter after	30 minutes

Local Policies/Security Options

Accounts

Policy	Setting
Accounts: Guest account status	Disabled

Administrative Templates

Policy definitions (ADMX files) retrieved from the local computer.

Windows Components/Windows Installer

Policy	Setting	Comment
Turn off Windows Installer	Enabled	
Disable Windows Installer		For non-managed applications only

User Configuration (Enabled)

Policies

Administrative Templates

Policy definitions (ADMX files) retrieved from the local computer.

Control Panel

sale

Scope	Details	Settings	Delegation
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User Configuration (Enabled)

Policies

Administrative Templates

Policy definitions (ADMX files) retrieved from the local computer.

Control Panel

Policy	Setting	Comment
Prohibit access to Control Panel and PC settings	Enabled	

System

Policy	Setting	Comment
Prevent access to the command prompt	Enabled	
Disable the command prompt script processing also?		No

System/Removable Storage Access

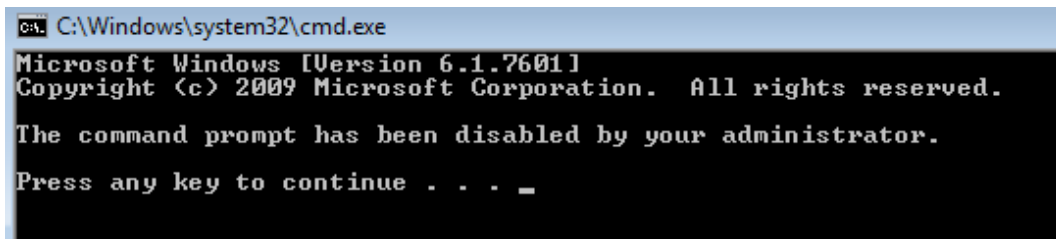
Policy	Setting	Comment
All Removable Storage classes: Deny all access	Enabled	
CD and DVD: Deny read access	Enabled	
CD and DVD: Deny write access	Enabled	
Floppy Drives: Deny read access	Enabled	
Floppy Drives: Deny write access	Enabled	
Removable Disks: Deny read access	Enabled	
Removable Disks: Deny write access	Enabled	
Set time (in seconds) to force reboot	Enabled	
Time (in seconds):		1800
Policy	Setting	Comment
Tape Drives: Deny read access	Enabled	
Tape Drives: Deny write access	Enabled	
WPD Devices: Deny read access	Enabled	
WPD Devices: Deny write access	Enabled	

Windows Components/Windows Update

Policy	Setting	Comment
Remove access to use all Windows Update features	Enabled	
Configure notifications:		0 - Do not show any notifications

Policy apply result in Client PC

CMD not work :-

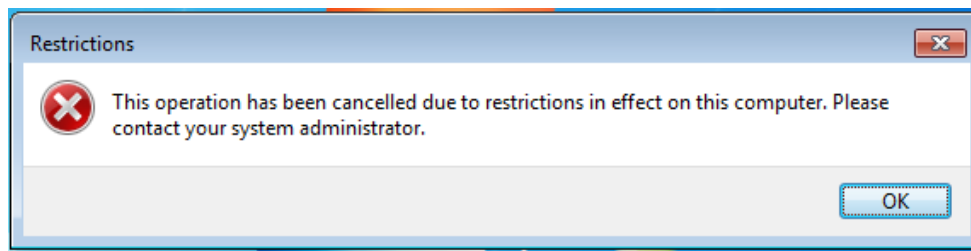


```
CA: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

The command prompt has been disabled by your administrator.

Press any key to continue . . . _
```

Other setting not open :-



Summary:

1. **AD DS Installation:** Installed and configured Active Directory Domain Services, creating a new domain (**Armour.exam**).
 2. **Organizational Unit Creation:** Created a new Organizational Unit (OU) called **Sales Department** and added users (**Sale1, Sale2, Sale3**).
 3. **Group Policy Creation:** Created and applied a **GPO** for the **Sales Department** with 10 security policies.
 4. **Client PC Domain Join:** Joined the client PC to the **Armour.exam** domain and tested the connection.
 5. **Group Policy Application:** Applied 10 policies successfully on the client PC using **gpupdate /force**.
-

Report on Hosting a Secure Website on IIS with HTTPS Enforcement

1. Introduction

- Objective: To host a secure website on IIS (Internet Information Services) with HTTPS enforcement using SSL certificates on Windows Server 2012.

2. Installing IIS on Windows Server 2012

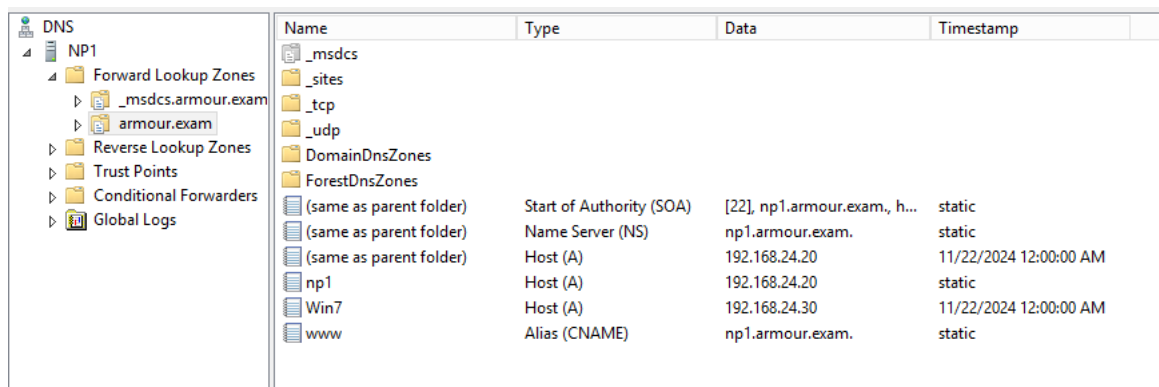
- Steps:
 - Opened **Server Manager**, selected **Add Roles and Features**.
 - Checked **Web Server (IIS)** under roles, selected necessary **role services** like **HTTP Redirection** and **WebDAV Publishing**.
 - Completed the installation of IIS and the required services.

3. Downloading and Uploading Website Files

- Downloaded HTML website templates from the internet.
- Placed the website files in the **C:\inetpub\wwwroot\site1** directory on the server.

4. Configuring DNS for Website Access

- Opened the **DNS Server** and navigated to **Forward Lookup Zones**.
- Created an **Alias (CNAME)** record for www.armour.exam, pointing to the **armour.exam** domain.



Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[22], np1.armour.exam., h...	static
(same as parent folder)	Name Server (NS)	np1.armour.exam.	static
(same as parent folder)	Host (A)	192.168.24.20	11/22/2024 12:00:00 AM
np1	Host (A)	192.168.24.20	static
Win7	Host (A)	192.168.24.30	11/22/2024 12:00:00 AM
www	Alias (CNAME)	np1.armour.exam.	static

5. Configuring SSL Certificate for HTTPS

- Opened **Internet Information Services (IIS) Manager**.
- Assigned a **self-signed SSL certificate** to the website to enable HTTPS.
- Installed the certificate to ensure secure browsing (HTTPS).

6. Creating and Configuring the Website in IIS

- Under **Sites** in IIS, clicked **Add Website**.
- Assigned the website name as **Site1**.
- Set the **Physical Path** to the folder containing the website files (**C:\inetpub\wwwroot\site1**).
- Configured the website **binding**:
 - Type: **HTTPS**
 - IP Address: **192.168.24.20**
 - Port: **443**
 - Hostname: www.armour.exam
- Selected the **SSL certificate** to be used for the site and clicked **OK**.

Add Website

Site name: Application pool:

Content Directory

Physical path:

Pass-through authentication

Binding

Type: IP address: Port:

Host name:

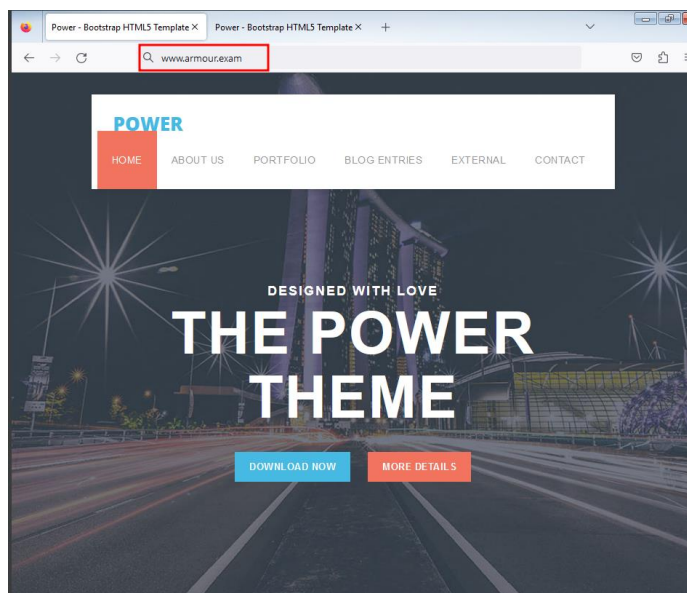
☐ Require Server Name Indication

SSL certificate:

☒ Start Website immediately

7. Testing the Website Access

- Opened the **browser** on the client PC.
- Entered the URL: <https://www.armour.exam>.
- Confirmed that the website was accessible and running securely with HTTPS.



8. Conclusion

- The website is now successfully hosted on IIS with HTTPS enforcement, using a self-signed SSL certificate.
- The **CNAME alias** was created in DNS to ensure the website is accessible by the www.armour.exam domain.
- The setup is now ready for secure access via HTTPS on client PCs within the network.